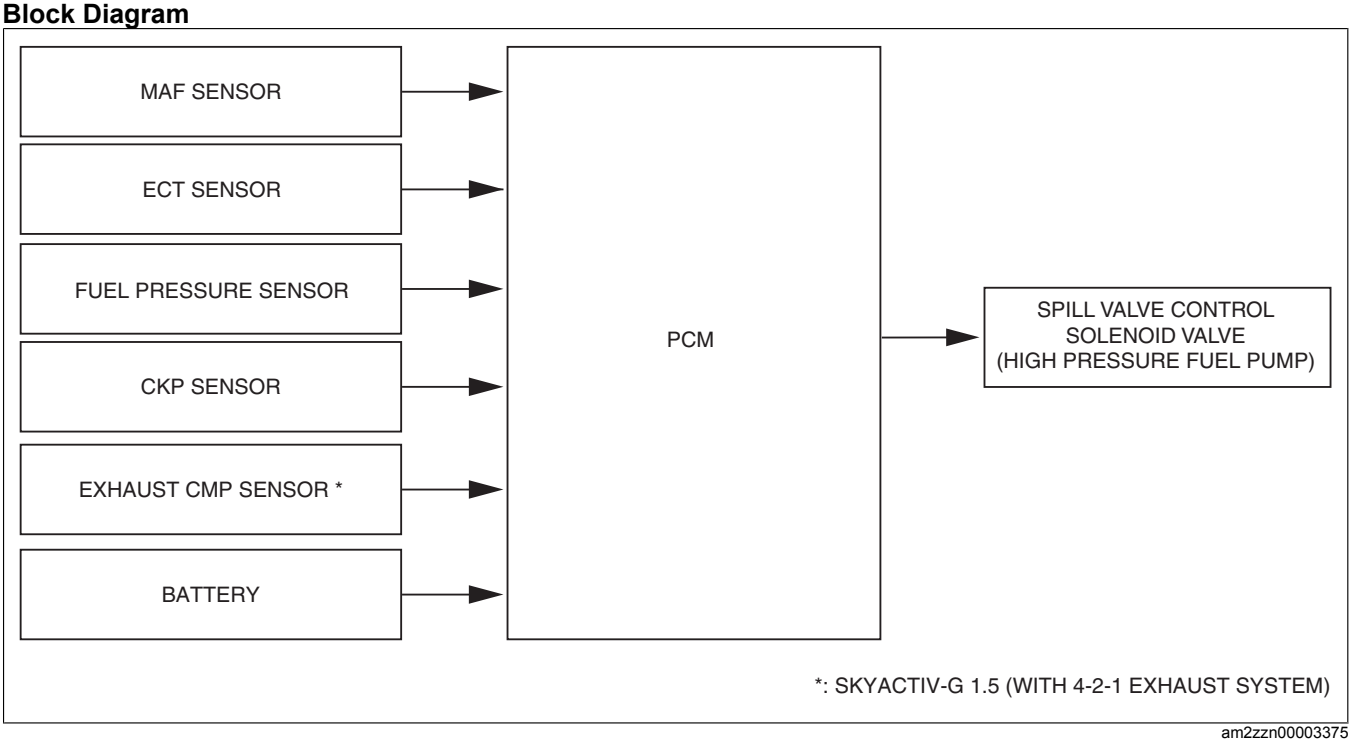
HIGH PRESSURE FUEL PUMP CONTROL [SKYACTIV-G 1.3, SKYACTIV-G 1.5]

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Outline

- Changes the fuel pressure applied to the fuel injector (according to engine operation conditions) to improve engine output and startability.
- The PCM determines the fuel pressure value corresponding to the engine operation conditions based on each input signal, and drives the spill valve control solenoid valve for optimum control of fuel pressure.

Block Diagram



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OPERATION

Directly after engine start

- During a cold engine start, the fuel pressure is raised to promote fuel atomization.

Basic Control

- The PCM determines the target fuel pressure according to charging efficiency and performs feedback control by monitoring the fuel pressure in the fuel distributor using the fuel pressure sensor.
  - When idling after engine warm up: **Approx. 3 MPa {31 kgf/cm^2, 435 psi}**
  - After engine warm up, engine speed is **3,000 rpm or more**, charging efficiency is **60 % or more**: **Approx. 15.0 MPa {153 kgf/cm^2, 2,176 psi}**